

Muhammad Ibrahim Umar

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Education

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

2024 – 2028

Bachelor's Degree, Artificial Intelligence (Sophomore – 2nd Year)

Topi, KPK, Pakistan

Technical Skills

Languages: C++, Python

AI / ML: PyTorch, scikit-learn, NumPy, Pandas, NLTK, SVM, K-Means, PCA, Linear Regression, Generative AI (Gemini API)

Web & APIs: FastAPI, Docker, Docker Compose

Data & Visualization: Plotly, Recharts, Altair, Seaborn, Jupyter Notebook, Streamlit

Frameworks & Tools: Qt Framework, SFML audio, PostgreSQL

Core Concepts: Data Structures and, Algorithms, OOP, Image Processing, Digital Logic Design, Software Documentation, SQL, Git, Git Hub

Projects

Philosophical Text Engine – Python, PyTorch, scikit-learn, FastAPI, Docker

- Built a hybrid ML pipeline to classify philosophical text passages sourced from Project Gutenberg into branches (e.g., Ethics) and specific schools of thought using a multi-tier Hierarchical MLP in PyTorch.
- Engineered a hybrid inference system combining a deterministic SVM baseline, probabilistic Neural Network predictions, and unsupervised K-Means clustering to deliver multi-perspective analysis of input passages.
- Applied TF-IDF vectorization (10,000 features) with NLTK preprocessing; used PCA to project passages into 2D space for visual analysis of philosophical “distance” between texts.
- Exposed the pipeline via a FastAPI **/predict** endpoint with Docker containerization; supports dual inference modes (Fast/Full-accuracy) with serialized model artifacts via Joblib.

Smart Saver – Grocery Substitution Engine – FastAPI, Gemini API, Docker | Micathon 2026

- Independently built a full-stack web application at Micathon 2026 that uses AI-powered OCR to scan grocery receipts, identify imported products, and recommend local alternatives to reduce consumer spending.
- Integrated Gemini 2.5 Flash Vision API on the FastAPI backend to extract item details from receipt images; cross-referenced results against a curated **substitutions.json** database for local brand matching.
- Prototyped a React 19 frontend with Recharts-powered interactive savings dashboards; implemented keyword categorization across 6 food groups and a surplus calculation engine.
- Containerized the full stack with Docker Compose; implemented a sample-receipt demo mode enabling zero-setup live demonstrations without physical receipts.

Anomaly Detection via Statistical Intensity Analysis – Python, NumPy, Pillow, Streamlit

- Developed an anomaly detection system using first-principles statistical analysis instead of Deep Learning, addressing the rare event problem in computer vision where limited defect datasets make Neural Networks difficult to train.
- Implemented a detection engine based on Mean and Standard Deviation; pixels flagged as anomalies when deviation exceeds threshold k . Developed isolated R/G/B channel maps to isolate specific defect types.
- Optimized to $O(N)$ pixel complexity for real-time performance on low-end hardware. Achieved zero-shot detection of structural cracks without pre-labeled training data and millisecond-range inference latency.
- Built an interactive Streamlit dashboard with a real-time k -value slider to adjust the detection threshold dynamically.

PC Parts Optimization Platform – FastAPI, Pandas

- Designing a multi-version budget-aware PC build recommendation platform; implements a heuristic scoring engine where users specify a budget and use case (Gaming, ML, General) to receive top GPU/CPU pairings.
- Engineered a weighted scoring system that normalizes raw hardware benchmarks (PassMark, clock speeds) via min-max scaling, applies use-case-specific weights, and adds under-budget headroom bonuses.
- Prototyped a React 18 + Tailwind frontend; FastAPI backend with Pandas for benchmark data ingestion. Future plans include web-scraped pricing, PostgreSQL, and a trained ML recommendation model.

Music Playlist Management System – C++, SFML, Data Structures and Algorithms

- Developed a high-performance music management and playback system without external databases, building a custom engine using advanced Data Structures and Algorithms to minimize memory overhead.
- Implemented AVL Trees for $O(\log n)$ lookups; a custom Polynomial Rolling Hash Table; and Merge Sort to rank Most Liked songs. Designed a Graph Theory/BFS recommendation engine based on user listening patterns.

Leadership & Experience

IET on Campus – GIKI

Sub Head (promoted from Volunteer)

Sep 2024 – Present

Topi, KPK, Pakistan

Shaukat Khanum Memorial Cancer Hospital & Research Centre

Volunteer – Ramazan Fundraising Drive

Ramazan 2026

Certifications

Business Communication (HP LIFE) | **Building with the Claude API** (Anthropic) | **Introduction to AI** (Google)